

RaceDay Event Management System

API Endpoint Plan

PROG6212 - PoE Part 1 | Author: Caleb Walsh

This document specifies every RESTful API endpoint the RaceDay system will expose. It covers Authentication, User Profile, Events, Categories, Event Enrolments, and Results, as required by the PoE Part 1 brief. Role values used below are: **None** (public, no login required), **Any** (any authenticated user), **Organiser**, and **Participant**.

1. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new User account as either an Organiser or a Participant.	None	{ fullName, email, password, role, phoneNumber, clubId }	201 Created; 400 Bad Request invalid data; 409 Conflict - email already registered
POST	/api/auth/login	Authenticates a registered User and returns a JWT access token containing their role.	None	{ email, password }	200 OK - returns token; 401 Unauthorized invalid credentials

2. User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Returns the profile of the currently logged-in User.	Any	None	200 OK; 401 Unauthorized
PUT	/api/users/me	Updates the profile details of the currently logged-in User.	Any	{ fullName, phoneNumber, clubId }	200 OK; 400 Bad Request; 401 Unauthorized

3. Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Returns a list of all upcoming RaceDay events.	None	None	200 OK
GET	/api/events/{id}	Returns the full details of a specific RaceDay event using the supplied event ID.	None	None	200 OK; 404 Not Found - event does not exist

POST	/api/events	Creates a new RaceDay event owned by the logged-in Organiser.	Organiser	{ eventName, description, eventDate, location, distance, eventType }	201 Created; 400 Bad Request; 401/403 - not an Organiser
HTTP Method	Route	Description	Role Required	Request Body	Expected Response
PUT	/api/events/{id}	Updates an existing event. Only the Organiser who created the event may update it.	Organiser	{ eventName, description, eventDate, location, distance, eventType }	200 OK; 400 Bad Request; 403 Forbidden - not the event owner; 404 Not Found
DELETE	/api/events/{id}	Deletes an event. Only the owning Organiser may delete it.	Organiser	None	204 No Content; 403 Forbidden; 404 Not Found

4. Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Returns all categories available for a specific event.	None	None	200 OK; 404 Not Found - event does not exist
POST	/api/events/{eventId}/categories	Creates a new category for an event. Only the owning Organiser may add categories.	Organiser	{ categoryName, minAge, maxParticipants, entryFee }	201 Created; 400 Bad Request; 403 Forbidden; 404 Not Found
PUT	/api/categories/{id}	Updates an existing category's details.	Organiser	{ categoryName, minAge, maxParticipants, entryFee }	200 OK; 400 Bad Request; 403 Forbidden; 404 Not Found
DELETE	/api/categories/{id}	Removes a category from an event, provided no Participants are enrolled in it.	Organiser	None	204 No Content; 403 Forbidden; 404 Not Found; 409 Conflict category has enrolments

5. Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
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POST	/api/events/{eventId}/enrolments	Enrols the logged-in Participant into an event under a chosen category.	Participant	{ categoryId }	201 Created enrolment record; 400 Bad Request; 404 Not Found; 409 Conflict already enrolled
GET	/api/users/me/enrolments	Returns all events the logged-in Participant has enrolled in.	Participant	None	200 OK; 401 Unauthorized
GET	/api/events/{eventId}/enrolments	Returns all Participants enrolled in a specific event. Only the owning Organisation may view this.	Organiser	None	200 OK; 403 Forbidden; 404 Not Found
DELETE	/api/enrolments/{id}	Cancels a Participant's own enrolment in an event.	Participant	None	204 No Content; 403 Forbidden not the enrolment owner; 404 Not Found

6. Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{enrolmentId}/result	Captures the finish time and finishing position for a Participant's enrolment. Only the event's Organisation may capture results.	Organiser	{ finishTime, position }	201 Created; 400 Bad Request; 403 Forbidden; 404 Not Found; 409 Conflict - result already captured
GET	/api/users/me/results	Returns the logged-in Participant's full personal performance history across all events.	Participant	None	200 OK; 401 Unauthorized
GET	/api/events/{eventId}/results	Returns all captured results for a specific event, e.g. for a results board.	Any	None	200 OK; 404 Not Found